Appendix B

Equation Registry and Era-Level Calibration

B.1 Canonical Equation: The Predatory Min-Max Function

The complete formal definition of the governing optimization algorithm is reproduced here as the canonical reference. All component equations in the manuscript derive from or refer back to this canonical equation.

The Predatory Min-Max Function — Canonical Entry

Full Set-Theoretic Hierarchy:[a] $E \subset P_{\text{uppet}} \subset I_{\text{buffer}} \subset I, O_{\text{racialized}} \cap I = \emptyset.$

Kernel Objective:[b]

$$\max_{\{P_i\}} \mathcal{E}(t) \quad \text{subject to} \quad M_{\text{eff}}(t) = M(t) - \lambda\, \Phi_{\text{load}}(t) < \tau \tag{B.1}$$

where the effective resistance variable is:[c]

$$M_{\text{eff}}(t) = M(t) - \lambda\, \Phi_{\text{load}}(t) \tag{B.2}$$

Suppression Envelope:[d]

$$\Sigma_{\text{sup}}(t) = \psi_s(t) + \psi_m(t) + R(t) + \Phi_{\text{load}}(t) \tag{B.3}$$

Crash Condition:[e]

$$\frac{dM}{dt} > \frac{d\Sigma_{\text{sup}}}{dt} \quad \Longleftrightarrow \quad M_{\text{eff}}(t) > \tau \tag{B.4}$$

Tier Benefit Ordering:[f]

$$\text{Benefit}(E) \gg \text{Benefit}(P_{\text{uppet}}) > \text{Benefit}(F_{\text{enforce}}) > \text{Benefit}(I_{\text{buffer}}) > \text{Benefit}(O_{\text{racialized}}) \tag{B.5}$$

Concession Invariant: $\forall\, R_i \in \mathcal{R}$ (non-kinetic reforms in the 1450–2026 dataset): $\Delta \max(R_i) = 0$

Suppression Allocation: $\psi_m(t) > 0$ only when $M(t) \to \tau$; sourced from increased $O_{\text{racialized}}$ extraction.

Boundary Dynamics: The reclassification operator $\mathcal{R}(x_i)$ defined in eq. (16.14) executes whenever $K(x_i) > K_{\text{tolerated}} \vee \text{comply}(x_i) = 0$, moving x_i from I_{buffer} to O_{final} with $\psi_s(x_i) \to 0$ instantaneously; empirical instantiations include Christiana 1851, Hamburg 1876, Mulford 1967, Ruby Ridge 1992, Waco 1993, and Perdi 2026 (§16.11).

Full definition and variable legend: Section 16.7.

[a]Registry restatement of eq. (16.5) (Chapter 10).

[b]Registry restatement of eq. (2.7) (Chapter 1) and eq. (16.6) (Chapter 10). Empirical validation: see the Antebellum Kernel Optimization case study (p. 49) and the Piketty-Saez-Zucman top-decile wealth concentration analysis. Tier 3 (structural); see the Empirical Methodology chapter (p. lxv).

[c]Registry restatement of eq. (16.7) (Chapter 10).

[d]Registry restatement of eq. (2.13) (Chapter 1) and eq. (16.8) (Chapter 10). Empirical validation: see the Post-1965 Backlash Wave case study (p. 66) and the ANES cross-racial solidarity proxy analysis. Tier 3 (structural); see the Empirical Methodology chapter (p. lxv).

[e]Registry restatement of eq. (2.14) (Chapter 1) and eq. (16.9) (Chapter 10). Empirical validation: Bacon's Rebellion (1676) and the Haitian Revolution (1791–1804) are the two documented cases where $M_{\text{eff}}(t) > \tau$ produced systemic kernel failure. Tier 3 (structural); see the Empirical Methodology chapter (p. lxv).

[f]Registry restatement of eq. (16.10) (Chapter 10). Empirical validation: Piketty-Saez-Zucman top-decile wealth concentration, Federal Reserve SCF median household wealth by quintile, and BJS incarceration data confirming $E \cap O_{\text{final}} = \emptyset$. Tier 3 (ordinal); see the Empirical Methodology chapter (p. lxv).

B.2 Electrodynamic Framework Registry

The following equations formalize the Unified Electrodynamic Framework (UEF) introduced in Chapter 0. They provide the hardware-layer translation of the set-theoretic and wave-physics formalisms deployed throughout the book. Their license to apply across substrates—the reason the same field equations govern both electrons in copper and humans in institutions—is the substrate-independence theorem established in Section 0.4.1: humans were executing these dynamics millennia before electromagnetism was formalized, and the electromagnetic vocabulary was borrowed from the prior social-power vocabulary. The registry below is therefore the formal apparatus the physicists left behind, applied to the system it was originally about.

Unified Electrodynamic Framework — Canonical Entry

Root Equation — The Unified Lorentz Force:

$$\vec{F}_{\text{total}} = \mathbf{Q}\,\vec{\mathcal{E}}_{\text{mat}} + \mathbf{Q}\left(\vec{v} \times \sum_{k=1}^{N} \rho_k \vec{B}_k\right) \tag{B.6}$$

where $\mathbf{Q}$ is the intersectional charge multivector, $\vec{\mathcal{E}}_{\text{mat}}$ is the material-wage electric field, and $\sum \rho_k \vec{B}_k$ is the superposition of N cultural magnetic fields. The cross product $\vec{v} \times \vec{B}$ is the mathematical signature of the psychological wage: always orthogonal to velocity, so it does zero work.[a]

Complex Suppression Allocation:

$$W = \psi_m + j\psi_s \tag{B.7}$$

where $\psi_m \in \mathbb{R}$ is the material wage (real power) and $j\psi_s \in j\mathbb{R}$ is the psychological wage (reactive power). In polar form: $W = R\,e^{j\theta}$ with $R = |W| = \sqrt{\psi_m^2 + \psi_s^2}$. The phase angle θ diagnoses the operating mode: $\theta = 90$ is the default (purely psychological); $\theta > 90$ is the fascism threshold (Quadrant II).[b]

Solidarity as Imaginary Cancellation:

$$W + W^* = 2\psi_m \tag{B.8}$$

When the Buffer Class adopts the complex conjugate $W^* = \psi_m - j\psi_s$ and aligns with the Out-group, the imaginary status wage cancels entirely and real material power doubles.[c]

Systemic Resistance (Drude Model):

$$R = \frac{m \cdot L}{n \cdot q^2 \cdot \tau \cdot A} \tag{B.9}$$

The Elite engineers high resistance for the Out-group by minimizing τ (mean free time between collisions). Stop-and-frisk, welfare audits, and algorithmic flags force individuals to collide with the state before building drift velocity.[d]

Enclosure Capacitance:

$$C = \frac{\epsilon A}{d} \tag{B.10}$$

where ϵ is the legal dielectric (property rights, inheritance law, FHA lending), A is plate area (geographic scope), and d is distance (redlining, segregation).[e]

Inductive Kickback:

$$V = -L\,\frac{di}{dt} \tag{B.11}$$

When reform attempts to break the oppressive current ($dt \to 0$), the inductor of cultural inertia discharges stored energy as fascist backlash.[f]

AC Complex Power and Real/Reactive Decomposition:

$$S = V \cdot I^* = |V|\,|I|\,e^{j\theta} = P_{\text{real}} + jQ_{\text{reactive}} \tag{B.12}$$

where $P_{\text{real}} = |V|\,|I|\cos\theta = \psi_m$ is the material wage (real power) and $Q_{\text{reactive}} = |V|\,|I|\sin\theta = \psi_s$ is the psychological wage (reactive power). DC mode is the $\theta = 0$ limit; fascism threshold is $\theta > 90$.[g]

N-Weighted Phase Current at the Elite Node:

$$I_E(t) = \sum_{k=1}^{N} \rho_k\, A_k\, \cos\!\big(\omega t + \phi_k\big) \tag{B.13}$$

The intersectional generalization of three-phase AC: N extraction axes weighted by ρ_k (population/coupling) with phases ϕ_k tuned by the interference engine. Kirchhoff's law at the Elite node enforces extraction-current conservation under axis dropout: suppression of one A_k is rebalanced by the others weighted by $\rho_k/(\sum_{m\neq k}\rho_m)$.[h]

Buffer-Class Work Theorem:

$$W^{\text{psych}}_{I_{\text{buffer}}}(t_0, t_1) = \int_{t_0}^{t_1} \mathbf{Q}\,(\vec{v} \times \vec{B}) \cdot \vec{v}\, d\tau \;\equiv\; 0 \tag{B.14}$$

The systemic work performed on the Buffer Class by the psychological wage is identically zero over any interval, because $\vec{v} \times \vec{B}$ is orthogonal to $\vec{v}$ by construction. Formal proof of Du Bois's "public and psychological wage" W.E.B Du Bois 1935.[i]

Reparations Integral:

$$\mathcal{R}(t_0, t_{\text{now}}) = \int_{t_0}^{t_{\text{now}}} \text{Re}\Big[V_{\text{state}}(\tau)\, I_O^*(\tau)\Big]\, d\tau \tag{B.15}$$

The time integral of real power flowing from the Out-group node to the Elite node over the historical operating interval of the extraction system. Lower limits are fixed by documented system-initialization dates ($t_0 = 1444$ Portuguese, $t_0 = 1492$ Hispaniola, $t_0 = 1619$ US). Three independent peer-reviewed calibrations exist: Craemer's compounded-labor estimate yields \$14–17 trillion Craemer 2015; Darity &

Mullen's wealth-gap-closure estimate yields \$10–12 trillion W. A. Darity and Mullen 2020; Coates' housing/education/credit survey provides the public articulation Ta-Nehisi Coates 2014. Tier 2 (multiple peer-reviewed calibrations of the same definite integral).[j]

Augmented Lagrangian Root Equation:

$$\mathcal{L}^*(q, \dot{q}, t) = T(\dot{q}, t) - V(q, t) - \mathcal{D}(q, \dot{q}, t) + \lambda\big(\tau - M_{\text{eff}}(t)\big) \tag{B.16}$$

where T is the mobilized social capacity (kinetic energy of the Out-group and Buffer class), V is the engineered constraint landscape (systemic diodes, redlining boundaries, Lyapunov ceilings), $\mathcal{D}$ is the Rayleigh dissipation function (bureaucratic friction, policing drag, administrative heat loss), and λ is the marginal cost of suppression (the Complex Wage $W = \psi_m + j\psi_s$). The constraint $M_{\text{eff}}(t) < \tau$ enforces the rebellion threshold.[k]

[a]Registry restatement of eq. (0.1) (Chapter 0). Tier 3 (ordinal/structural); see the Empirical Methodology chapter (p. lxv).
[b]Registry restatement of eq. (3.5) (Chapter 1). Tier 3 (ordinal/structural).
[c]Registry restatement of eq. (3.8) (Chapter 1). Tier 3 (ordinal/structural).
[d]Registry restatement of eq. (0.8) (Chapter 0). Tier 3 (ordinal/structural).
[e]Registry restatement of eq. (0.9) (Chapter 0). Tier 3 (ordinal/structural).
[f]Registry restatement of eq. (0.10) (Chapter 0). Tier 3 (ordinal/structural).
[g]Registry restatement of eqs. (0.13)–(0.14) (Chapter 0). Tier 3 (ordinal/structural).
[h]Registry restatement of eqs. (0.15)–(0.16) (Chapter 0). Tier 3 (ordinal/structural).
[i]Registry restatement of eq. (0.19) (Chapter 0). Tier 3 (structural/derivational).
[j]Registry restatement of eq. (0.20) (Chapter 0). Tier 2 (multiple independent peer-reviewed calibrations).
[k]Registry restatement of eq. (1.2) (Chapter 1). Tier 3 (ordinal/structural); the augmented Lagrangian formalizes the directional claim that the Elite optimize extraction subject to a rebellion-threshold constraint. See the Empirical Methodology chapter (p. lxv).

B.3 Canonical Symbol Registry

The manuscript uses a dual representation: kernel semantics (`Max/Min`) and dynamic state variables. The registry below is canonical.

Symbol	Type	Definition	Primary Mapping
$\mathcal{E}(t)$	Dynamic objective	Time-indexed extraction output routed toward E.	Dynamic analogue of max

Symbol	Type	Definition	Primary Mapping
$M(t)$	Dynamic state	Time-indexed class-coherence threat to kernel stability.	Dynamic analogue of min stress
τ	Threshold	Domestic collapse threshold for coherent class solidarity.	Failure boundary for $M(t)$ / $\lvert S_{\text{total}}\rvert$
$\psi(t) = \psi_s(t) + \psi_m(t)$	Control variable	Suppression allocation to I_{buffer}: ψ_s (status wage, always active) + ψ_m (material wage, ≥ 0; deployed when $M(t) \to \tau$; sourced from increased $O_{\text{racialized}}$ extraction).	Suppression envelope component
$R(t)$	Control variable	Kinetic repression capacity via F_{enforce}.	Suppression envelope component
$\phi_{k,j}$	Phase variable	Axis-level phase shift for axis k and subgroup j.	Intersectional interference component
Φ_j	Phase aggregate	Compound subgroup phase shift: $\Phi_j = \sum_k \phi_{k,j}$.	Subgroup wave displacement
$\Phi_{\text{load}}(t)$	Dispersion index	Aggregate phase-dispersion load across active subgroups.	Suppression envelope component
$S_{\text{total}}(t)$	Wave state	Net class-solidarity signal under intersectional phase loading.	Collapse test against τ
ρ_τ	Diagnostic ratio	Proximity index $\rho_\tau = M(t)/\tau$.	Runtime-log risk indicator
$K(x_i)$	Individual state	Kinetic capacity of individual x_i: armament $\times$ demonstrated willingness to deploy against F_{enforce}.	Reclassification input; §16.11

Symbol	Type	Definition	Primary Mapping
$K_{\text{tolerated}}$	Threshold	E's threshold for permitted Buffer-Class kinetic capacity under the current runtime; era-calibrated.	Reclassification threshold; §16.11
$\mathcal{R}(x_i)$	Boundary operator	Reclassification operator: conditional assignment of x_i to I_{buffer} or O_{final} based on $(K(x_i), K_{\text{tolerated}}, \text{comply}(x_i))$; see eq. (16.14).	Dynamic boundary function on eq. (16.13)
X_{temporal}	Mechanism set	Temporal Enclosure: the set of all sovereign debt instruments $D_{\text{sovereign}}$ whose issuance securitizes $L_{\text{future}}(O \cup I_{\text{buffer}})$ and transfers the discounted future labor surplus to E. See eq. (16.16).	Post-kinetic extraction vector; replaces spatial/kinetic enclosures as primary mechanism in mature architectures.
L_{future}	Resource variable	Aggregate future labor capacity of $O \cup I_{\text{buffer}}$: the productive output not yet performed, whose present value is securitized by $D_{\text{sovereign}}$.	Extraction target of X_{temporal}.
$D_{\text{sovereign}}$	Stock variable	Sovereign debt outstanding: the total accumulated issuance of X_{temporal} instruments; triggers eq. (20.15) when $D_{\text{sovereign}} > \bar{D}$.	Primary instrument of X_{temporal}; accumulation clock for Interface Swap.

Symbol	Type	Definition	Primary Mapping
$\delta_E(t)$	Extraction rate	Annual extraction increment from financial repression: $\delta_E = \pi(t) - i(t) > 0$ when $r_{\text{real}} < 0$; the real purchasing power transferred per unit time from I_{buffer}/O to E. See eq. (16.17).	Stealth extraction rate; operates without kinetic enforcement.
$\mathcal{E}_{X_{\text{temporal}}}(t)$	Dynamic objective	Instantaneous temporal extraction rate: $\frac{dD_{\text{sovereign}}}{dt} \cdot \delta_E(t) \cdot \mathbf{1}[r < g]$. See eq. (16.18).	Component of $\mathcal{E}(t)$ active under $r < g$ condition; scales with sovereign debt accumulation.
$\mathcal{A}$	Architecture variable	Extraction architecture: the specific legal, monetary, and institutional substrate on which the Predatory Min-Max Function executes at a given historical moment ($\mathcal{A}_{\text{old}}$, $\mathcal{A}_{\text{new}}$). See eq. (20.16).	Argument of the Polymorphic Reboot Operator.
Δ_E	Continuity invariant	Change in E's asset base across an Interface Swap: $\Delta_E = \text{Assets}(E, \mathcal{A}_{\text{new}}) - \text{Assets}(E, \mathcal{A}_{\text{old}}) \geq 0$. See eq. (20.17).	Hard constraint on every Polymorphic Interface Swap; $\Delta_E < 0$ falsifies the reboot operator.

Symbol	Type	Definition	Primary Mapping
$\mathcal{L}_{\text{Dalio}}$	Cycle index	Dalio Empire Index: 8-factor composite measuring national power (education, innovation, competitiveness, output, trade share, military, financial center, reserve currency), normalized $[0, 1]$. See eq. (16.21).	Macro-clock for the 250-year empire cycle; decline below $\mathcal{L}_{\text{Dalio}} \approx 0.75$ correlates with Demographic Paradox Limit activation.

Non-reuse constraint: E denotes the Elite class set only; $\mathcal{E}(t)$ denotes the extraction output function. ψ denotes psychological wage only. All phase operations use ϕ/Φ notation only. The symbol $\mathcal{R}$ is disambiguated by arity: $\mathcal{R}$ without arguments (as in the Concession Invariant, §B) denotes the set of non-kinetic reforms in the 1450–2026 dataset; $\mathcal{R}(x_i)$ with an individual argument denotes the reclassification operator defined in equation (16.14). The two occupy orthogonal mathematical categories (set vs. function) and cannot be confused in well-formed expressions.

B.4 Era-Level Interference Calibration Matrix

The following ranges are bounded internal calibration intervals on a normalized [0,1] scale. They calibrate comparability across eras at a normalized scale without asserting fine-grained precision.

Era / Runtime	Φ_{load}	ρ_τ	Confidence Tier	Evidence Pathway
1486 (Lisbon)	[0.10, 0.20]	[0.70, 0.85] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 2	Chapter 3 runtime + initial partition architecture + psychological wage initialization W.E.B Du Bois 1935.

Era / Runtime	$\Phi_{\mathbf{load}}$	ρ_τ	**Confidence Tier**	**Evidence Pathway**
1676–1787	[0.15, 0.55]	(1.00, 0.75]; **Anchor** ($\rho_\tau = 1.0$; see p. lxv)	Tier 1	**Anchor** ($\rho_\tau = 1.0$): Bacon crash evidence + legal codification sequence + constitutional patch dynamics.
1704–1865	[0.45, 0.60]	[0.45, 0.65] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 2	Enforcement-class consolidation + slave patrol and 13th Amendment continuity evidence.
Gendered axis	[0.35, 0.55]	[0.40, 0.60] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 2	Chapter 8 runtime + coverture + reproductive criminalization evidence.
1870s–1960s	[0.55, 0.72]	[0.50, 0.68] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 1	Redlining containment + two-party scaling + capture dynamics Rothstein 2017; T.-N Coates 2014.
1968–Present	[0.75, 0.92]	[0.80, 0.98] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 1	Variable Swap + carceral expansion + agenda-setting and elite-policy evidence Gilens, M. and Page, B.I 2014.

Era / Runtime	$\Phi_{\mathbf{load}}$	ρ_τ	**Confidence Tier**	**Evidence Pathway**
2007–2012 (Great Recession Recompile)	[0.78, 0.90]	[0.82, 0.96] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 1	Housing crash + racialized wealth loss + bottom-half balance-sheet floor + asset recovery preserving $\Delta \max = 0$ Board of Governors of the Federal Reserve System 2026; Congressional Budget Office 2024; Kochhar and Cilluffo 2017; Pfeffer, Danziger, and Schoeni 2013.
1992–1995 (Ruby Ridge / Waco / OKC)	[0.70, 0.85]	[0.72, 0.88] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 1	First live-action reclassification of Buffer-Class subjects under rewritten ROE; Patriot/militia misroute of kinetic response U. D. o. Justice 1994; O. o. P. R. U.S. Department of Justice 2001; Danforth 2000; Treasury 1993; Center 2015; Pitcavage 2001.
Disarmament timeline	[0.60, 0.85]	[0.70, 0.95] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 2	Kinetic-control sequence as adjunct to interference suppression across periods.

Era / Runtime	$\Phi_{\mathbf{load}}$	ρ_τ	**Confidence Tier**	**Evidence Pathway**
Prescriptive stress test	[0.82, 0.95]	[0.92, 1.05] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 2	Reform paradox and concession dynamics under near-threshold conditions.
Global scaling	[0.50, 0.90]	[0.55, 0.95] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 3	Imperial containment variance by region and bloc coordination UN High Commissioner for Human Rights 2026.
Final output	[0.85, 0.97]	[0.95, 1.08] (calibrated against Bacon/Haiti anchors; see p. lxv)	Tier 2	Book-level synthesis of terminal saturation and diminishing suppression returns.

Confidence tiers: Tier 1 = directly reported or transparently derivable from a peer-reviewed source or public dataset, with no undisclosed analytical step; Tier 2 = public dataset with disclosed author operationalisation; Tier 3 = ordinal or structural claim with no quantitative calibration attempted and its basis stated.